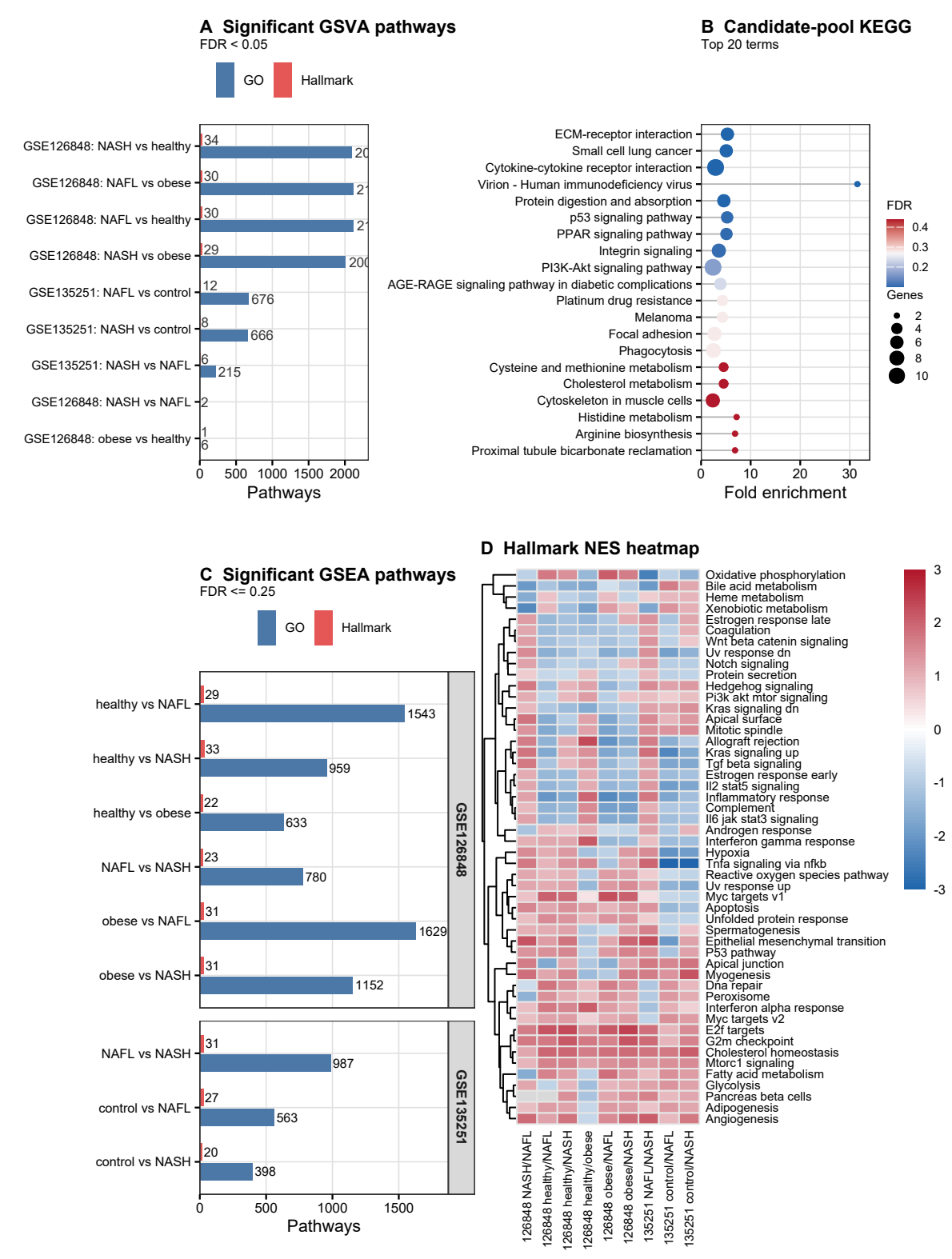

A Significant GSVA pathways
FDR < 0.05
GO
Hallmark
GSE126848: NASH vs healthy
GSE126848: NAFL vs obese
GSE126848: NAFL vs healthy
GSE126848: NASH vs obese
GSE135251: NAFL vs control
GSE135251: NASH vs control
GSE135251: NASH vs NAFL
GSE126848: NASH vs NAFL
GSE126848: obese vs healthy
34
30
30
29
12
676
8
666
6
215
2
1
6
0
500
1000
1500
2000
Pathways
B Candidate-pool KEGG
Top 20 terms
ECM-receptor interaction
Small cell lung cancer
Cytokine-cytokine receptor interaction
Virion - Human immunodeficiency virus
Protein digestion and absorption
p53 signaling pathway
PPAR signaling pathway
Integrin signaling
PI3K-Akt signaling pathway
AGE-RAGE signaling pathway in diabetic complications
Platinum drug resistance
Melanoma
Focal adhesion
Phagocytosis
Cysteine and methionine metabolism
Cholesterol metabolism
Cytoskeleton in muscle cells
Histidine metabolism
Arginine biosynthesis
Proximal tubule bicarbonate reclamation
0
10
20
30
Fold enrichment
FDR
0.4
0.3
0.2
Genes
2
4
6
8
10
C Significant GSEA pathways
FDR <= 0.25
GO
Hallmark
healthy vs NAFL
healthy vs NASH
healthy vs obese
NAFL vs NASH
obese vs NAFL
obese vs NASH
29
1543
33
959
22
633
23
780
31
1629
31
1152
GSE126848
NAFL vs NASH
control vs NAFL
control vs NASH
31
987
27
563
20
398
GSE135251
0
500
1000
1500
Pathways
D Hallmark NES heatmap
Oxidative phosphorylation
Bile acid metabolism
Heme metabolism
Xenobiotic metabolism
Estrogen response late
Coagulation
Wnt beta catenin signaling
Uv response dn
Notch signaling
Protein secretion
Hedgehog signaling
Pi3k akt mtor signaling
Kras signaling dn
Apical surface
Mitotic spindle
Allograft rejection
Kras signaling up
Tgf beta signaling
Estrogen response early
Il2 stat5 signaling
Inflammatory response
Complement
Il6 jak stat3 signaling
Androgen response
Interferon gamma response
Hypoxia
Tnfa signaling via nfkb
Reactive oxygen species pathway
Uv response up
Myc targets v1
Apoptosis
Unfolded protein response
Spermatogenesis
Epithelial mesenchymal transition
P53 pathway
Apical junction
Myogenesis
Dna repair
Peroxisome
Interferon alpha response
Myc targets v2
E2f targets
G2m checkpoint
Cholesterol homeostasis
Mtorc1 signaling
Fatty acid metabolism
Glycolysis
Pancreas beta cells
Adipogenesis
Angiogenesis
126848 NASH/NAFL
126848 healthy/NAFL
126848 healthy/NASH
126848 healthy/obese
126848 obese/NAFL
126848 obese/NASH
135251 NAFL/NASH
135251 control/NAFL
135251 control/NASH
3
2
1
0
-1
-2
-3